



BICYCLE DRIVE ASSISTANT - REPORT

CUSTOMER
ESTIMATED START OF PRODUCTION
ANALYSIS COMPLETED BY

PROGRAM/PROJECT
ESTIMATED YEARLY VOLUME
ANALYSIS DATE

BILL OF MATERIALS

GATES	PRODUCT LINE	TOOTH COUNT	MOUNTING INTERFACE	PRODUCT DESCRIPTION	MATERIAL	FINISH/COLOR
DRIVE 1						
CRANKSET	CDX	50	24 mm Spindle	FC550 170BM 50CDX-0/45.5 NG	Forged Alloy	Black
REAR SPROCKET	CDX	20	Rohloff Splined	XF1120RSMN	Stainless	Silver
BELT	CDX	115	/	11M-115T-12CT BLACK	Polyurethane	Black

BELT DRIVE COMPARISON

GATES	DRIVE 1
BIKE	
APPLICATION	Urban Sportive
REAR WHEEL SIZE	700c/29er
REAR TIRE SIZE	38 mm
REAR WHEEL DIAMETER (MM)	698.00
TYPE OF BIKE	Pedal Bike
EBIKE CLASS	/
EBIKE DRIVE MANUFACTURER	/
EBIKE DRIVE MODEL	/
EBIKE DRIVE SUBMODEL	/
GEAR	
GEARING TYPE	Internal Gear Hub
BRAND	Rohloff
O.L.D. (MM)	135
MAIN MODEL	SpeedHUB
SUBMODEL	135 O.L.D.



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BELT DRIVE COMPARISON (CONTINUED)

GATES	DRIVE 1
SPROCKET	
FRONT SPROCKET ASSEMBLY CHOICE	Crankset Assembly
CRANKSET MOUNTING INTERFACE	24 mm Spindle
CRANKSET MODEL	S550
TOOTH COUNT	50
CRANKSET GUARD SELECTION	No Guard
CRANKSET BELTLINE	54.7
CRANK ARM LENGTH	170
CRANKSET COLOR	Black
REAR SPROCKET PRODUCT LINE	CDX
REAR SPROCKET MOUNTING TYPE	Rohloff Splined
REAR SPROCKET OFFSET TYPE	Standard
REAR SPROCKET ORIENTATION	Branding Outboard
REAR SPROCKET SIZE (T)	20
BELT	
BELT PRODUCT LINE	CDX
REAR CENTER DISTANCE MIN (MM)	/
REAR CENTER DISTANCE MAX (MM)	/
BELT SIZE (T)	115
BELT LINE (MM)	54.7
TENSIONING IDLER (3RD PULLEY) PLANNED?	False



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GATES	DRIVE 1	
BELT TOOTH COUNT	115	
CENTER DISTANCE (MM)	436.84	
TAKE-UP ADJUSTMENT FOR TENSION (MM)	438.84	
BELT LINE (MM)	54.7	
EBIKE DRIVE MODEL	/	
GEARING MANUFACTURER	Rohloff	
GEARING MODEL	SpeedHUB	
SPROCKETS (FRONT/REAR)	50	20
GEARING RATIO (MIN/MAX)	0.279	1.467

SPEED		CADENCE (RPM)
KPH	MPH	
70	43	
67.5	42	
65	40	
62.5	39	
60	37	
57.5	36	
55	34	
52.5	33	109
50	31	104
47.5	30	98
45	28	93
42.5	26	88
40	25	83
37.5	23	78
35	22	73
32	20	66
30	19	62
27.5	17	57
25	16	52
22.5	14	47
20	12	41
17.5	11	36
15	9	31
12.5	8	
10	6	109
7.5	5	82
5	3	54
2.5	2	
0	0	

LOWEST STABLE 2-WHEELER SPEED 30
RPM MINIMUM

This cadence/speed analysis helps to ensure the bicycle being designed will have sufficient low gearing for climbing, as well as comfortable rider pedal cadence at higher speeds. This is especially important for eBikes, where the assistance stops at particular speeds, and is where the rider will spend a majority of their time while at steady state higher speeds.

The most up-to-date product information is in the [Gates Belt Drive Binder](#)

Further application or product-specific details can be found in our [Technical Bulletins](#)

Additional instructions can be found in the [Bicycle Drive Assistant Video](#)